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Microprocessors – Microcontrollers

Laboratory Report 1

LED Animation

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1. Introduction

This report presents the implementation of Laboratory 1 (LED) of the Microprocessors – Microcontrollers course, using the STM32F103C6 microcontroller simulated in Proteus 8 Professional and programmed with STM32CubeIDE (HAL library).

The laboratory is divided into two independent hardware setups, each reused across several exercises:

- Exercises 1–5: a 4-way traffic-light system built from 12 LEDs (RED / YELLOW / GREEN for each of the North, South, East and West directions) together with one 7-segment display (7SEG-COM-ANODE) used to show a countdown value.
- Exercises 6–10: a 12-LED analog clock, where each LED represents one of the 12 hour marks on a clock face, used to display the current hour, minute and second.

For every LED used in this laboratory, the negative (cathode) pin is connected to a GPIO pin of the STM32, while the positive (anode) pin is tied to 3.3 V. As a consequence, all LEDs in this design are active-low: driving the GPIO pin to logic 0 (GPIO_PIN_RESET) turns the LED on, while driving it to logic 1 (GPIO_PIN_SET) turns the LED off. The same active-low principle applies to the segments of the common-anode 7-segment display.

2. Hardware Design

2.1 Schematic for Exercises 1–5: 4-Way Traffic Light & 7-Segment Display

Figure 2.1 shows the Proteus schematic shared by Exercises 1 to 5. Twelve LEDs (three colors × four directions) are connected to pins PA4–PA15 of the STM32F103C6, and one common-anode 7-segment display is connected to pins PB0–PB6 (segments A–G), used to display the countdown value of the traffic light cycle.

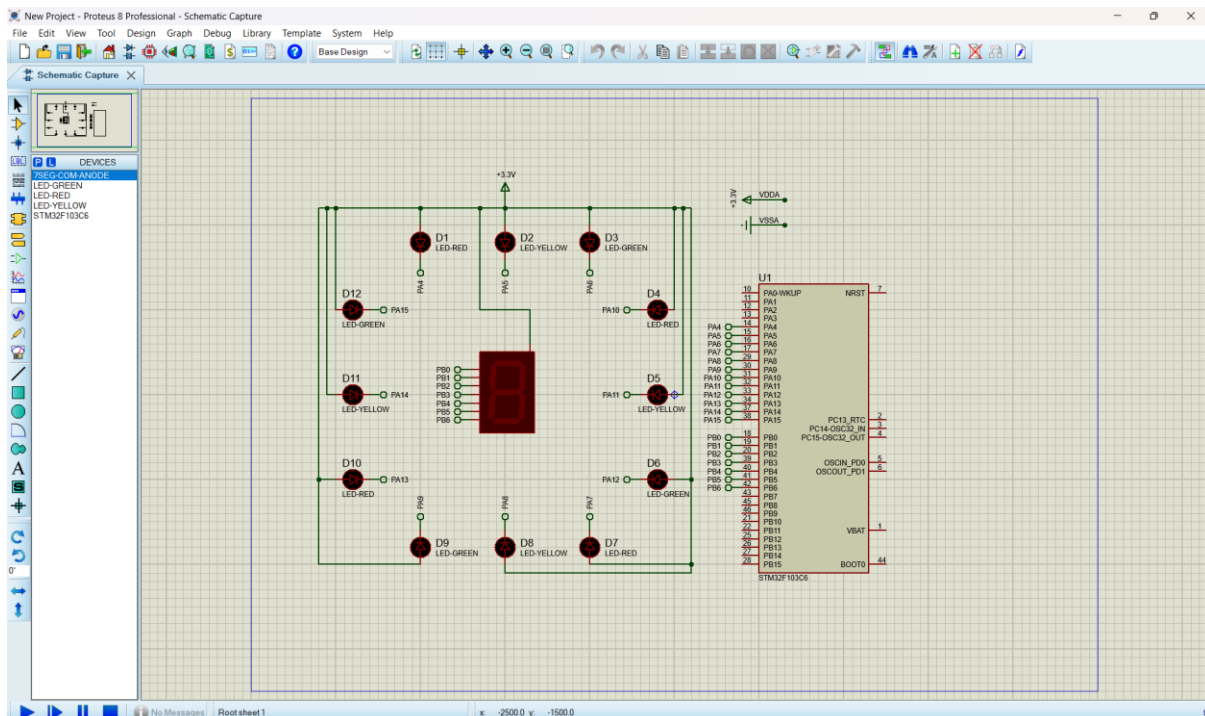


Figure 2.1: Proteus schematic for Exercises 1–5 (traffic light and 7-segment display)

[mcu-mpu_lab_schematic/lab1_ex1-5.png](#) at main · garrettnhan-coder/mcu-mpu_lab_schematic

2.2 Schematic for Exercises 6–10: 12-LED Analog Clock

Figure 2.2 shows the Proteus schematic shared by Exercises 6 to 10. Twelve LEDs are connected to pins PA4–PA15, arranged to represent the 12 hour marks of an analog clock face.

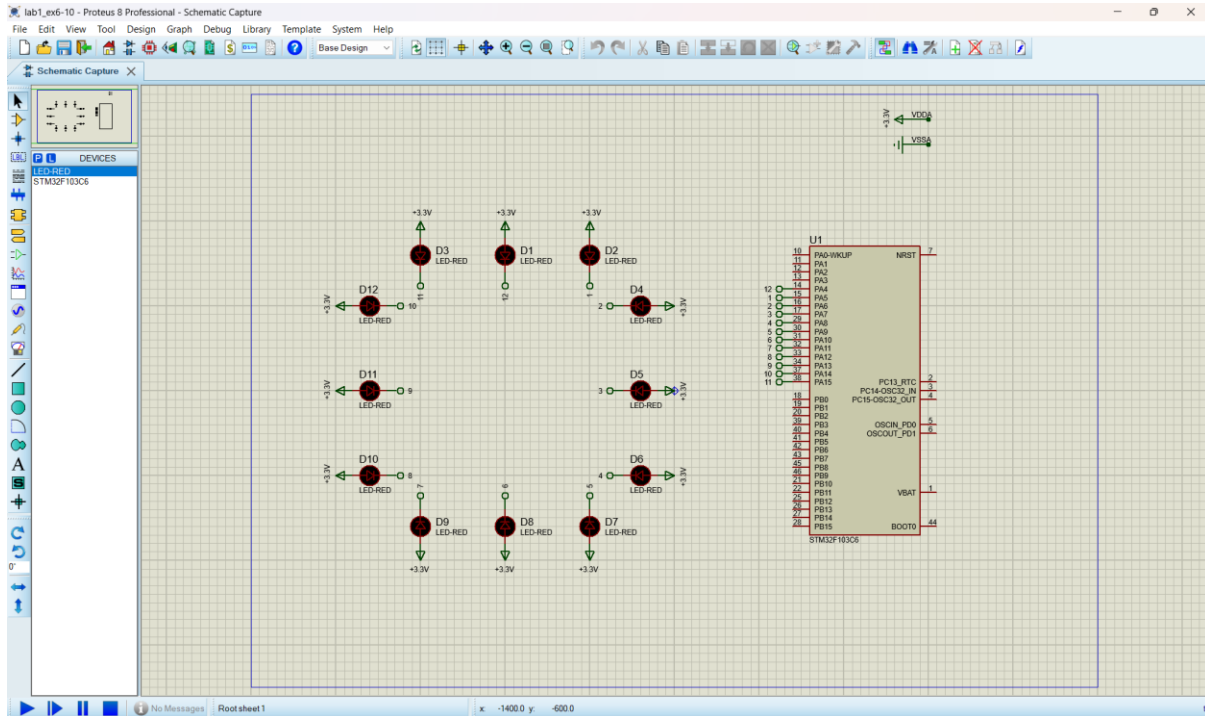


Figure 2.2: Proteus schematic for Exercises 6–10 (12-LED analog clock).

[mcu-mpu_lab_schematic/lab1_ex6-10.png](#) at main · garrettnhan-coder/mcu-mpu_lab_schematic

3. Exercise 1: Two-LED Alternating States

Using the schematic in Figure 2.1, two LEDs of the North traffic light (N_RED and N_YELLOW) are toggled so that their states switch every 2 seconds: when N_RED is on, N_YELLOW is off, and vice versa.

Listing 3.1 — Exercise 1: Alternating two LEDs every 2 seconds

```
while (1) {
    HAL_GPIO_WritePin(N_RED_GPIO_Port, N_RED_Pin, SET);
    HAL_GPIO_WritePin(N_YELLOW_GPIO_Port, N_YELLOW_Pin, RESET);
    HAL_Delay(2000);

    HAL_GPIO_WritePin(N_RED_GPIO_Port, N_RED_Pin, RESET);
    HAL_GPIO_WritePin(N_YELLOW_GPIO_Port, N_YELLOW_Pin, SET);
    HAL_Delay(2000);
}
```

4. Exercise 2: Single-Junction Traffic Light

Exercise 1 is extended by adding a third LED, N_GREEN, to form a complete RED–YELLOW–GREEN traffic-light cycle for a single direction (North). The timing of one cycle is 5 seconds for RED, 2 seconds for YELLOW and 3 seconds for GREEN.

Listing 4.1 — Exercise 2: Single-junction traffic light cycle (5-2-3 seconds)

```
while (1) {
    HAL_GPIO_WritePin(N_RED_GPIO_Port, N_RED_Pin, RESET);
    HAL_GPIO_WritePin(N_YELLOW_GPIO_Port, N_YELLOW_Pin, SET);
    HAL_GPIO_WritePin(N_GREEN_GPIO_Port, N_GREEN_Pin, SET);
```

```

    HAL_Delay(5000);

    HAL_GPIO_WritePin(N_RED_GPIO_Port, N_RED_Pin, SET);
    HAL_GPIO_WritePin(N_YELLOW_GPIO_Port, N_YELLOW_Pin, RESET);
    HAL_GPIO_WritePin(N_GREEN_GPIO_Port, N_GREEN_Pin, SET);
    HAL_Delay(2000);

    HAL_GPIO_WritePin(N_YELLOW_GPIO_Port, N_YELLOW_Pin, SET);
    HAL_GPIO_WritePin(N_YELLOW_GPIO_Port, N_YELLOW_Pin, SET);
    HAL_GPIO_WritePin(N_GREEN_GPIO_Port, N_GREEN_Pin, RESET);
    HAL_Delay(3000);
}

```

5. Exercise 3: 4-Way Traffic Light Extension

The single-junction design of Exercise 2 is extended to a full 4-way intersection. Twelve LEDs (RED, YELLOW and GREEN for each of the North, South, East and West directions) are arranged around the 7-segment display as shown previously in Figure 2.1, reproducing the reference layout of a 4-way traffic light. Each signal is wired to one GPIO pin in the range PA4–PA15 and labeled accordingly in STM32CubeMX (e.g. N_RED_Pin, S_GREEN_Pin, W_YELLOW_Pin, etc.). No additional source code is required for this exercise; the full timing logic that drives all twelve LEDs together is implemented as part of the integration in Exercise 5.

6. Exercise 4: 7-Segment Display Function `display7SEG(int num)`

The function `display7SEG(int num)` accepts an input from 0 to 9 and lights up the corresponding segments (A–G) of the common-anode 7-segment display to show that digit. Since the display is common-anode, a segment is turned on by driving its pin to RESET (logic 0).

Listing 6.1 — Exercise 4: `display7SEG(int num)` function

```

void display7SEG(int num)
{
    switch (num) {
        case 0:
            HAL_GPIO_WritePin(SEG_A_GPIO_Port, SEG_A_Pin, RESET);
            HAL_GPIO_WritePin(SEG_B_GPIO_Port, SEG_B_Pin, RESET);
            HAL_GPIO_WritePin(SEG_C_GPIO_Port, SEG_C_Pin, RESET);
            HAL_GPIO_WritePin(SEG_D_GPIO_Port, SEG_D_Pin, RESET);
            HAL_GPIO_WritePin(SEG_E_GPIO_Port, SEG_E_Pin, RESET);
            HAL_GPIO_WritePin(SEG_F_GPIO_Port, SEG_F_Pin, RESET);
            HAL_GPIO_WritePin(SEG_G_GPIO_Port, SEG_G_Pin, SET);
            break;
        case 1:
            HAL_GPIO_WritePin(SEG_A_GPIO_Port, SEG_A_Pin, SET);
            HAL_GPIO_WritePin(SEG_B_GPIO_Port, SEG_B_Pin, RESET);
            HAL_GPIO_WritePin(SEG_C_GPIO_Port, SEG_C_Pin, RESET);
            HAL_GPIO_WritePin(SEG_D_GPIO_Port, SEG_D_Pin, SET);
            HAL_GPIO_WritePin(SEG_E_GPIO_Port, SEG_E_Pin, SET);
            HAL_GPIO_WritePin(SEG_F_GPIO_Port, SEG_F_Pin, SET);
            HAL_GPIO_WritePin(SEG_G_GPIO_Port, SEG_G_Pin, SET);
            break;
        case 2:
            HAL_GPIO_WritePin(SEG_A_GPIO_Port, SEG_A_Pin, RESET);
            HAL_GPIO_WritePin(SEG_B_GPIO_Port, SEG_B_Pin, RESET);
            HAL_GPIO_WritePin(SEG_C_GPIO_Port, SEG_C_Pin, SET);
            HAL_GPIO_WritePin(SEG_D_GPIO_Port, SEG_D_Pin, RESET);
            HAL_GPIO_WritePin(SEG_E_GPIO_Port, SEG_E_Pin, RESET);
            HAL_GPIO_WritePin(SEG_F_GPIO_Port, SEG_F_Pin, SET);
            HAL_GPIO_WritePin(SEG_G_GPIO_Port, SEG_G_Pin, RESET);
            break;
        case 3:

```

```

        HAL_GPIO_WritePin(SEG_A_GPIO_Port, SEG_A_Pin, RESET);
        HAL_GPIO_WritePin(SEG_B_GPIO_Port, SEG_B_Pin, RESET);
        HAL_GPIO_WritePin(SEG_C_GPIO_Port, SEG_C_Pin, RESET);
        HAL_GPIO_WritePin(SEG_D_GPIO_Port, SEG_D_Pin, RESET);
        HAL_GPIO_WritePin(SEG_E_GPIO_Port, SEG_E_Pin, SET);
        HAL_GPIO_WritePin(SEG_F_GPIO_Port, SEG_F_Pin, SET);
        HAL_GPIO_WritePin(SEG_G_GPIO_Port, SEG_G_Pin, RESET);
        break;
    case 4:
        HAL_GPIO_WritePin(SEG_A_GPIO_Port, SEG_A_Pin, SET);
        HAL_GPIO_WritePin(SEG_B_GPIO_Port, SEG_B_Pin, RESET);
        HAL_GPIO_WritePin(SEG_C_GPIO_Port, SEG_C_Pin, RESET);
        HAL_GPIO_WritePin(SEG_D_GPIO_Port, SEG_D_Pin, SET);
        HAL_GPIO_WritePin(SEG_E_GPIO_Port, SEG_E_Pin, SET);
        HAL_GPIO_WritePin(SEG_F_GPIO_Port, SEG_F_Pin, RESET);
        HAL_GPIO_WritePin(SEG_G_GPIO_Port, SEG_G_Pin, RESET);
        break;
    case 5:
        HAL_GPIO_WritePin(SEG_A_GPIO_Port, SEG_A_Pin, RESET);
        HAL_GPIO_WritePin(SEG_B_GPIO_Port, SEG_B_Pin, SET);
        HAL_GPIO_WritePin(SEG_C_GPIO_Port, SEG_C_Pin, RESET);
        HAL_GPIO_WritePin(SEG_D_GPIO_Port, SEG_D_Pin, RESET);
        HAL_GPIO_WritePin(SEG_E_GPIO_Port, SEG_E_Pin, SET);
        HAL_GPIO_WritePin(SEG_F_GPIO_Port, SEG_F_Pin, RESET);
        HAL_GPIO_WritePin(SEG_G_GPIO_Port, SEG_G_Pin, RESET);
        break;
    case 6:
        HAL_GPIO_WritePin(SEG_A_GPIO_Port, SEG_A_Pin, RESET);
        HAL_GPIO_WritePin(SEG_B_GPIO_Port, SEG_B_Pin, SET);
        HAL_GPIO_WritePin(SEG_C_GPIO_Port, SEG_C_Pin, RESET);
        HAL_GPIO_WritePin(SEG_D_GPIO_Port, SEG_D_Pin, RESET);
        HAL_GPIO_WritePin(SEG_E_GPIO_Port, SEG_E_Pin, RESET);
        HAL_GPIO_WritePin(SEG_F_GPIO_Port, SEG_F_Pin, RESET);
        HAL_GPIO_WritePin(SEG_G_GPIO_Port, SEG_G_Pin, RESET);
        break;
    case 7:
        HAL_GPIO_WritePin(SEG_A_GPIO_Port, SEG_A_Pin, RESET);
        HAL_GPIO_WritePin(SEG_B_GPIO_Port, SEG_B_Pin, RESET);
        HAL_GPIO_WritePin(SEG_C_GPIO_Port, SEG_C_Pin, RESET);
        HAL_GPIO_WritePin(SEG_D_GPIO_Port, SEG_D_Pin, SET);
        HAL_GPIO_WritePin(SEG_E_GPIO_Port, SEG_E_Pin, SET);
        HAL_GPIO_WritePin(SEG_F_GPIO_Port, SEG_F_Pin, SET);
        HAL_GPIO_WritePin(SEG_G_GPIO_Port, SEG_G_Pin, SET);
        break;
    case 8:
        HAL_GPIO_WritePin(SEG_A_GPIO_Port, SEG_A_Pin, RESET);
        HAL_GPIO_WritePin(SEG_B_GPIO_Port, SEG_B_Pin, RESET);
        HAL_GPIO_WritePin(SEG_C_GPIO_Port, SEG_C_Pin, RESET);
        HAL_GPIO_WritePin(SEG_D_GPIO_Port, SEG_D_Pin, RESET);
        HAL_GPIO_WritePin(SEG_E_GPIO_Port, SEG_E_Pin, RESET);
        HAL_GPIO_WritePin(SEG_F_GPIO_Port, SEG_F_Pin, RESET);
        HAL_GPIO_WritePin(SEG_G_GPIO_Port, SEG_G_Pin, RESET);
        break;
    case 9:
        HAL_GPIO_WritePin(SEG_A_GPIO_Port, SEG_A_Pin, RESET);
        HAL_GPIO_WritePin(SEG_B_GPIO_Port, SEG_B_Pin, RESET);
        HAL_GPIO_WritePin(SEG_C_GPIO_Port, SEG_C_Pin, RESET);
        HAL_GPIO_WritePin(SEG_D_GPIO_Port, SEG_D_Pin, RESET);
        HAL_GPIO_WritePin(SEG_E_GPIO_Port, SEG_E_Pin, SET);
        HAL_GPIO_WritePin(SEG_F_GPIO_Port, SEG_F_Pin, RESET);
        HAL_GPIO_WritePin(SEG_G_GPIO_Port, SEG_G_Pin, RESET);
        break;
    }
}

```

To test the function on its own, it is invoked from an infinite loop that counts from 0 to 9 repeatedly, with one second between each digit:

Listing 6.2 — Exercise 4: Testing display7SEG()

```
int counter = 0;
while (1) {
    if (counter >= 10) counter = 0;
    display7SEG(counter++);
    HAL_Delay(1000);
}
```

7. Exercise 5: Full Integration – 4-Way Traffic Light with Countdown

Finally, the 4-way traffic light of Exercise 3 and the display7SEG() function of Exercise 4 are integrated into a single system: the 7-segment display continuously counts down from 9 to 0, and the twelve traffic-light LEDs change state according to the count, reproducing a synchronized intersection cycle:

- Count 9–7 (3 s): West–East direction runs (GREEN), North–South stops (RED).
- Count 6–5 (2 s): West–East slows down (YELLOW), North–South remains stopped (RED).
- Count 4–2 (3 s): North–South runs (GREEN), West–East stops (RED).
- Count 1–0 (2 s): North–South slows down (YELLOW), West–East remains stopped (RED).

Listing 7.1 — Exercise 5: Full 4-way traffic light integrated with the countdown display

```
while (1)
{
    // 9-8-7: W-E RUN, N-S STOP
    HAL_GPIO_WritePin(N_RED_GPIO_Port, N_RED_Pin, RESET);
    HAL_GPIO_WritePin(N_YELLOW_GPIO_Port, N_YELLOW_Pin, SET);
    HAL_GPIO_WritePin(N_GREEN_GPIO_Port, N_GREEN_Pin, SET);
    HAL_GPIO_WritePin(S_RED_GPIO_Port, S_RED_Pin, RESET);
    HAL_GPIO_WritePin(S_YELLOW_GPIO_Port, S_YELLOW_Pin, SET);
    HAL_GPIO_WritePin(S_GREEN_GPIO_Port, S_GREEN_Pin, SET);

    HAL_GPIO_WritePin(W_RED_GPIO_Port, W_RED_Pin, SET);
    HAL_GPIO_WritePin(W_YELLOW_GPIO_Port, W_YELLOW_Pin, SET);
    HAL_GPIO_WritePin(W_GREEN_GPIO_Port, W_GREEN_Pin, RESET);
    HAL_GPIO_WritePin(E_RED_GPIO_Port, E_RED_Pin, SET);
    HAL_GPIO_WritePin(E_YELLOW_GPIO_Port, E_YELLOW_Pin, SET);
    HAL_GPIO_WritePin(E_GREEN_GPIO_Port, E_GREEN_Pin, RESET);

    for (int i = 9; i >= 7; i--) { display7SEG(i); HAL_Delay(1000); }

    // 6-5: W-E SLOW, N-S STOP
    HAL_GPIO_WritePin(N_RED_GPIO_Port, N_RED_Pin, RESET);
    HAL_GPIO_WritePin(N_YELLOW_GPIO_Port, N_YELLOW_Pin, SET);
    HAL_GPIO_WritePin(N_GREEN_GPIO_Port, N_GREEN_Pin, SET);
    HAL_GPIO_WritePin(S_RED_GPIO_Port, S_RED_Pin, RESET);
    HAL_GPIO_WritePin(S_YELLOW_GPIO_Port, S_YELLOW_Pin, SET);
    HAL_GPIO_WritePin(S_GREEN_GPIO_Port, S_GREEN_Pin, SET);

    HAL_GPIO_WritePin(W_RED_GPIO_Port, W_RED_Pin, SET);
    HAL_GPIO_WritePin(W_YELLOW_GPIO_Port, W_YELLOW_Pin, RESET);
    HAL_GPIO_WritePin(W_GREEN_GPIO_Port, W_GREEN_Pin, SET);
    HAL_GPIO_WritePin(E_RED_GPIO_Port, E_RED_Pin, SET);
    HAL_GPIO_WritePin(E_YELLOW_GPIO_Port, E_YELLOW_Pin, RESET);
    HAL_GPIO_WritePin(E_GREEN_GPIO_Port, E_GREEN_Pin, SET);

    for (int i = 6; i >= 5; i--) { display7SEG(i); HAL_Delay(1000); }
```

```

// 4-3-2: W-E STOP, N-S RUN
HAL_GPIO_WritePin(N_RED_GPIO_Port, N_RED_Pin, SET);
HAL_GPIO_WritePin(N_YELLOW_GPIO_Port, N_YELLOW_Pin, SET);
HAL_GPIO_WritePin(N_GREEN_GPIO_Port, N_GREEN_Pin, RESET);
HAL_GPIO_WritePin(S_RED_GPIO_Port, S_RED_Pin, SET);
HAL_GPIO_WritePin(S_YELLOW_GPIO_Port, S_YELLOW_Pin, SET);
HAL_GPIO_WritePin(S_GREEN_GPIO_Port, S_GREEN_Pin, RESET);

HAL_GPIO_WritePin(W_RED_GPIO_Port, W_RED_Pin, RESET);
HAL_GPIO_WritePin(W_YELLOW_GPIO_Port, W_YELLOW_Pin, SET);
HAL_GPIO_WritePin(W_GREEN_GPIO_Port, W_GREEN_Pin, SET);
HAL_GPIO_WritePin(E_RED_GPIO_Port, E_RED_Pin, RESET);
HAL_GPIO_WritePin(E_YELLOW_GPIO_Port, E_YELLOW_Pin, SET);
HAL_GPIO_WritePin(E_GREEN_GPIO_Port, E_GREEN_Pin, SET);

for (int i = 4; i >= 2; i--) { display7SEG(i); HAL_Delay(1000); }

// 1-0: W-E STOP, N-S SLOW
HAL_GPIO_WritePin(N_RED_GPIO_Port, N_RED_Pin, SET);
HAL_GPIO_WritePin(N_YELLOW_GPIO_Port, N_YELLOW_Pin, RESET);
HAL_GPIO_WritePin(N_GREEN_GPIO_Port, N_GREEN_Pin, SET);
HAL_GPIO_WritePin(S_RED_GPIO_Port, S_RED_Pin, SET);
HAL_GPIO_WritePin(S_YELLOW_GPIO_Port, S_YELLOW_Pin, RESET);
HAL_GPIO_WritePin(S_GREEN_GPIO_Port, S_GREEN_Pin, SET);

HAL_GPIO_WritePin(W_RED_GPIO_Port, W_RED_Pin, RESET);
HAL_GPIO_WritePin(W_YELLOW_GPIO_Port, W_YELLOW_Pin, SET);
HAL_GPIO_WritePin(W_GREEN_GPIO_Port, W_GREEN_Pin, SET);
HAL_GPIO_WritePin(E_RED_GPIO_Port, E_RED_Pin, RESET);
HAL_GPIO_WritePin(E_YELLOW_GPIO_Port, E_YELLOW_Pin, SET);
HAL_GPIO_WritePin(E_GREEN_GPIO_Port, E_GREEN_Pin, SET);

for (int i = 1; i >= 0; i--) { display7SEG(i); HAL_Delay(1000); }
}

```

8. Exercise 6: Schematic Design and LED Test Program (Analog Clock)

Twelve LEDs are connected to pins PA4–PA15 of the STM32F103C6, as shown in Figure 2.2, and labeled in STM32CubeMX as LED_12, LED_1, LED_2, ..., LED_11, following the layout of an analog clock face. To verify that every LED is wired correctly, all LEDs are first turned off, and then a single LED is lit at a time, moving sequentially around the clock face every second.

Listing 8.1 — Exercise 6: Sequential LED test program

```

int i = 0;
while (1) {
    clearAllClock();
    setNumberOnClock(i);
    HAL_Delay(1000);
    i++;
    if (i >= 12) i = 0;
}

```

At the start of every iteration, `clearAllClock()` switches all 12 LEDs off; then `setNumberOnClock(i)` lights up only the LED at position `i`. This confirms that each of the 12 connections functions correctly.

9. Exercise 7: `clearAllClock()` Function

The function `clearAllClock()` turns off all 12 LEDs of the clock at once, by calling `clearNumberOnClock()` (see Section 11) for every position from 0 to 11.

Listing 9.1 — Exercise 7: `clearAllClock()` function

```
void clearAllClock(void) {
    for (int i = 0; i < 12; i++) {
        clearNumberOnClock(i);
    }
}
```

10. Exercise 8: `setNumberOnClock(int num)` Function

The function `setNumberOnClock(int num)` accepts a position from 0 to 11 (where 0 corresponds to the 12 o'clock position) and turns on the LED at that position by driving the corresponding pin to logic 0 (active-low).

Listing 10.1 — Exercise 8: `setNumberOnClock(int num)` function

```
void setNumberOnClock(int num) {
    switch (num) {
        case 0: HAL_GPIO_WritePin(LED_12_GPIO_Port, LED_12_Pin, RESET); break;
        case 1: HAL_GPIO_WritePin(LED_1_GPIO_Port, LED_1_Pin, RESET); break;
        case 2: HAL_GPIO_WritePin(LED_2_GPIO_Port, LED_2_Pin, RESET); break;
        case 3: HAL_GPIO_WritePin(LED_3_GPIO_Port, LED_3_Pin, RESET); break;
        case 4: HAL_GPIO_WritePin(LED_4_GPIO_Port, LED_4_Pin, RESET); break;
        case 5: HAL_GPIO_WritePin(LED_5_GPIO_Port, LED_5_Pin, RESET); break;
        case 6: HAL_GPIO_WritePin(LED_6_GPIO_Port, LED_6_Pin, RESET); break;
        case 7: HAL_GPIO_WritePin(LED_7_GPIO_Port, LED_7_Pin, RESET); break;
        case 8: HAL_GPIO_WritePin(LED_8_GPIO_Port, LED_8_Pin, RESET); break;
        case 9: HAL_GPIO_WritePin(LED_9_GPIO_Port, LED_9_Pin, RESET); break;
        case 10: HAL_GPIO_WritePin(LED_10_GPIO_Port, LED_10_Pin, RESET); break;
        case 11: HAL_GPIO_WritePin(LED_11_GPIO_Port, LED_11_Pin, RESET); break;
    }
}
```

11. Exercise 9: `clearNumberOnClock(int num)` Function

Symmetrically, `clearNumberOnClock(int num)` accepts a position from 0 to 11 and turns off the LED at that position by driving the corresponding pin to logic 1.

Listing 11.1 — Exercise 9: `clearNumberOnClock(int num)` function

```
void clearNumberOnClock(int num) {
    switch (num) {
        case 0: HAL_GPIO_WritePin(LED_12_GPIO_Port, LED_12_Pin, SET); break;
        case 1: HAL_GPIO_WritePin(LED_1_GPIO_Port, LED_1_Pin, SET); break;
        case 2: HAL_GPIO_WritePin(LED_2_GPIO_Port, LED_2_Pin, SET); break;
        case 3: HAL_GPIO_WritePin(LED_3_GPIO_Port, LED_3_Pin, SET); break;
        case 4: HAL_GPIO_WritePin(LED_4_GPIO_Port, LED_4_Pin, SET); break;
        case 5: HAL_GPIO_WritePin(LED_5_GPIO_Port, LED_5_Pin, SET); break;
        case 6: HAL_GPIO_WritePin(LED_6_GPIO_Port, LED_6_Pin, SET); break;
        case 7: HAL_GPIO_WritePin(LED_7_GPIO_Port, LED_7_Pin, SET); break;
        case 8: HAL_GPIO_WritePin(LED_8_GPIO_Port, LED_8_Pin, SET); break;
        case 9: HAL_GPIO_WritePin(LED_9_GPIO_Port, LED_9_Pin, SET); break;
        case 10: HAL_GPIO_WritePin(LED_10_GPIO_Port, LED_10_Pin, SET); break;
        case 11: HAL_GPIO_WritePin(LED_11_GPIO_Port, LED_11_Pin, SET); break;
    }
}
```

12. Exercise 10: Full Analog Clock Integration

The whole system is finally integrated to display a running clock. At any given moment, only three LEDs are turned on, one each for the current hour, minute and second. Since the clock face only has

12 positions, the minute and second values (0–59) are scaled down to the range 0–11 by dividing by 5.

Listing 12.1 — Exercise 10: Full analog clock (hour / minute / second)

```
int hour = 1, minute = 15, second = 45;

clearAllClock();
while (1) {
    clearAllClock();
    setNumberOnClock(hour % 12);
    setNumberOnClock(minute / 5);
    setNumberOnClock(second / 5);

    HAL_Delay(1000);

    second++;
    if (second >= 60) { second = 0; minute++; }
    if (minute >= 60) { minute = 0; hour++; }
    if (hour >= 12) hour = 0;
}
```